

COCOMO.py

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1 # COCOMO Model Effort and Cost Estimation
2
3 def calculate_effort_and_cost(size, a, b, cost_per_person_month):
4     effort = a * (size ** b)
5     cost = effort * cost_per_person_month
6     print(f"\nEffort required: {effort:.2f} person-months")
7     print(f"Estimated cost: ${cost:.2f}")
8
9 if __name__ == "__main__":
10     models = {1: ("Organic", 2.4, 1.05), 2: ("Semi-Detached", 3.0, 1.12), 3: ("Embedded", 3.6, 1.20)}
11     try:
12         model_type = int(input("Select the COCOMO model type (1-Organic, 2-Semi-Detached, 3-Embedded): "))
13         if model_type not in models:
14             print("Invalid model type selected.")
15             exit()
16
17         model_name, a, b = models[model_type]
18         print(f"\nSelected model type: {model_name}")
19
20         size = float(input("Enter the size of the project in KLOC (thousands of lines of code): "))
21         cost_per_person_month = float(input("Enter the cost per person-month: "))
22         calculate_effort_and_cost(size, a, b, cost_per_person_month)
23     except ValueError:
24         print("Invalid input. Please enter numeric values.")
```